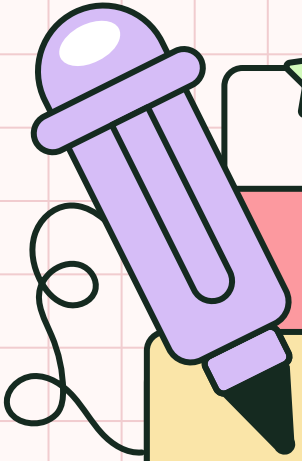
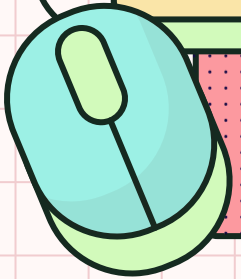
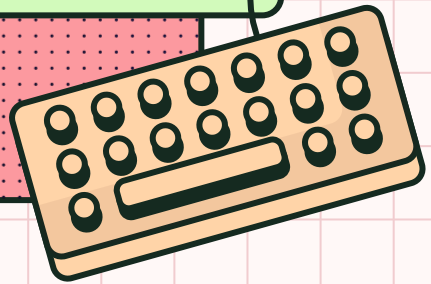
LESSON 2
GIRLS WHO CODE

CONDITIONAL STATEMENTS

AGENDA:

1. ICEBREAKER
2. WOMAN IN TECH
3. LESSON
4. CODING EXERCISE
5. REFLECTION

ICEBREAKER!

Imagine you're stuck on a **deserted island...**

- Pick 3 things you'd want to bring with you.
- Write them down and be ready to share.
- Then, you'll say your pick out loud and explain why you chose your item.

WOMEN IN TECH

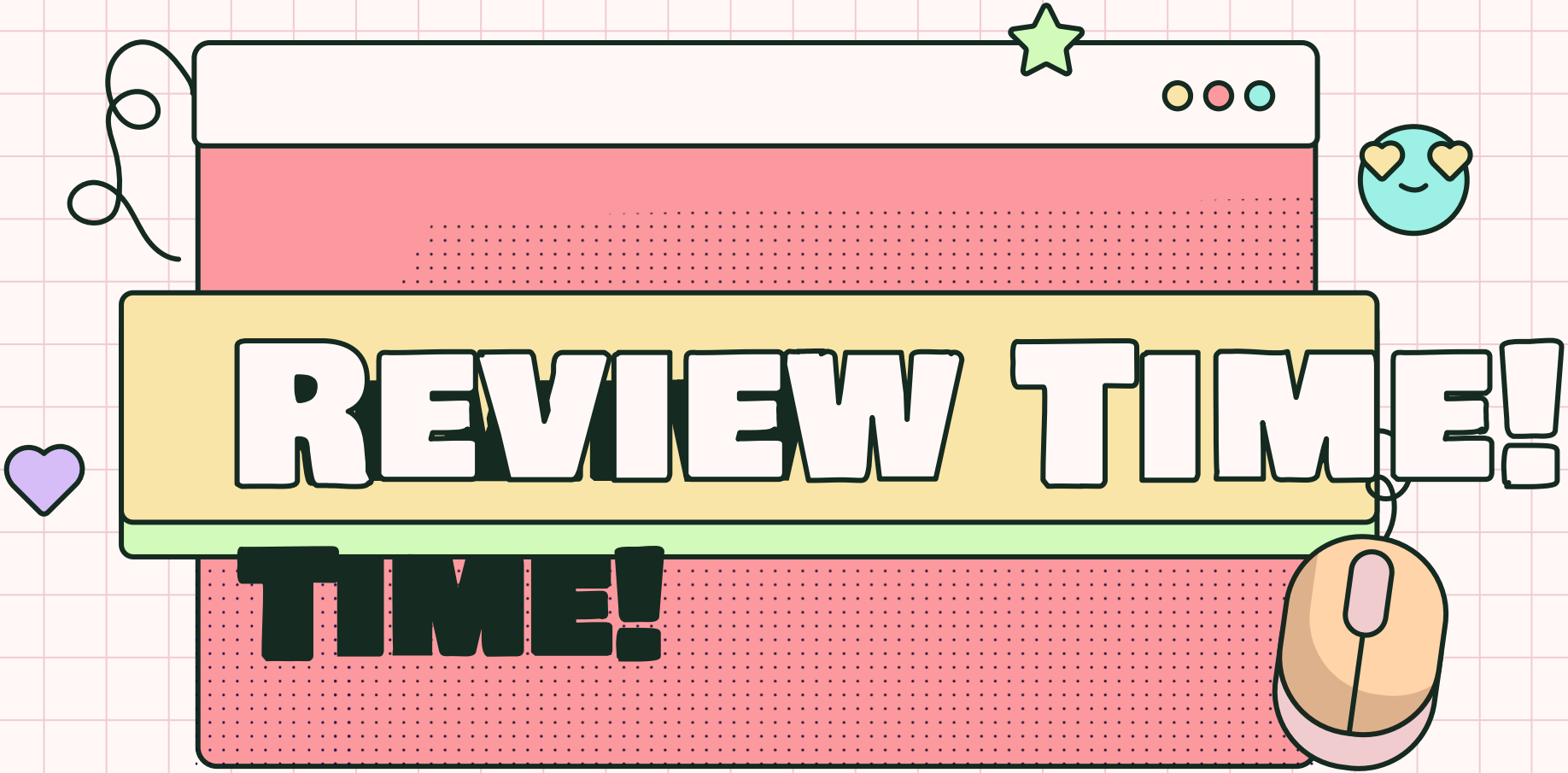


ADA LOVELACE

DISCUSS...

- What subjects did Ada Lovelace study besides mathematics?
- What kinds of things did Ada's notes on the analytical machine describe?
- Which of Ada's character traits are most inspiring to you?

REVIEW TIME!



Math/Logical Operators

Let's review the material we went over last week!

Addition:	'+' Plus sign
Subtraction:	'-' Minus sign
Multiplication:	'*' Multiplication sign
Division:	'/' Division symbol
Modulo:	'%' Tells us the remainder after dividing two numbers
Boolean operators:	NOT- this negates (gives us the opposite of) a boolean. Greater than(>), Less than(<), Equal to (==) ALWAYS USE TWO '='! We only use one (=) when creating a variable!

DATA TYPES

INTEGER

A whole number
(Ex: 1, 15, -25, 81, 5000000, -424

DOUBLE

A decimal number of any length
(Ex: 5.2, -8.9889394)

FLOAT

A less precise, more compact double
(Ex: 0.5, 7.3, 1.5)

CHAR

Short for character, a single key!
(Ex, t , A, $)

STRING

A word/phrase surrounded by quotes
(Ex: “hello” “hey”)

BOOL

True or False

LET'S PRACTICE!

- False
- “6.2324”
- “hello girls who code!”
- 7
- &
- 92

ANSWERS

- False # bool
- “6.2324” # string
- “hello girls who code!” # string
- 7 # int
- & # char
- 0.23 # float

VARIABLES

CREATE VARIABLES

x = 29
name = “Rachel”

CHANGE TYPE

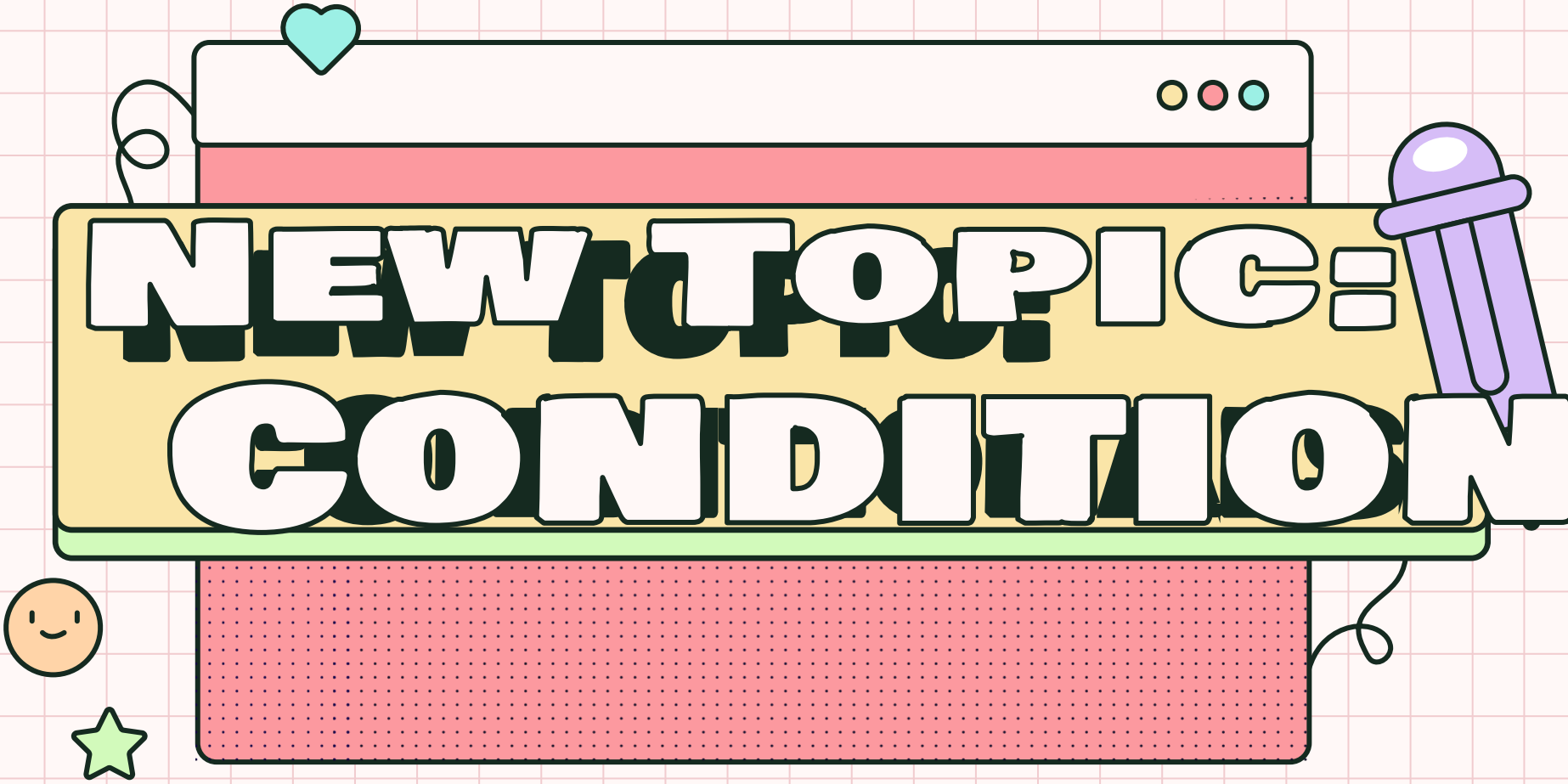
x = 9
name = “Anna”

CASTING

x = str(3), y = int(12)
z = float(2.5)

GET THE TYPE

x = 5, y = “Emma”
print(type(x))


NEW TOPIC:
CONDITION

What are Conditionals?

If some method is true, then do something...

If:



Then:



...else, do another action.

How does this look in Python?

IF:

```
if expression_1:
    statement_1
```

ELIF:

```
if expression_1:
    statement_1
elif expression_2:
    statement_2
```

ELSE:

```
if expression_1:
    statement_1
elif expression_2:
    statement_2
else:
    statement_3
```

Let's Practice!

Input:

```
if 3+2 >= 4:
    print("5 >= 4 is true!")
```

Output:

What will the input print out?

5 >= 4 is true!

Let's Practice!

Input:

```
if 3+2 < 4:
    print("5 >= 4 is true!")
```

Output:

What will the input print out?

Nothing!

Let's Practice!

Input:

```
if 3 > 4:
    print("3 > 4 is true!")
elif: "hello" == "hello":
    print("hello is true!")
```

Output:

What will the input print out?

hello is true!

Let's Practice!

Input:

```
if 3 < 4:
    print("3 < 4 is true!")
elif: "hello" == "hello":
    print("hello is true!")
```

Output:

What will the input print out?

3 < 4 is true!

Let's Practice!

Input:

```
if 3 > 4:
    print("3 > 4 is true!")
elif: "hello" == "goodbye":
    print("hello is true!")
else:
    print("nothing is true!")
```

Output:

What will the input print out?

nothing is true!

CODING TIME

https://tinyurl.com/gwclecture2

ANSWERS

```
if height >= 48:
    print("Enjoy the ride!")
else:
    print("Sorry, you're not
tall enough yet.")
```

ANSWERS

```
if age < 13:
    print("Ticket is $8")
elif age <= 64:
    print("Ticket is $14")
else:
    print("Ticket is $10")
```

ANSWERS

Output: “Perfect weather for a walk!”

ANSWERS

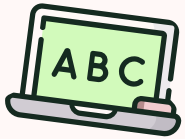
```
if favorite_snack == "popcorn":
    print("Movie night ready!")
else:
    print("Time to raid the pantry!")
```

Let's reflect:

What did you learn today?



What did you find challenging?



What do you feel most comfortable doing?



THAT'S IT FOR THIS WEEK!

We will practice more conditionals throughout the next lessons!

